

Pick Up The Ball

Special Aspects of mobile autonomous Systems

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Agenda

- ❑ Task & NAO in detail
- ❑ Object detection & interaction
- ❑ Problems
- ❑ Result

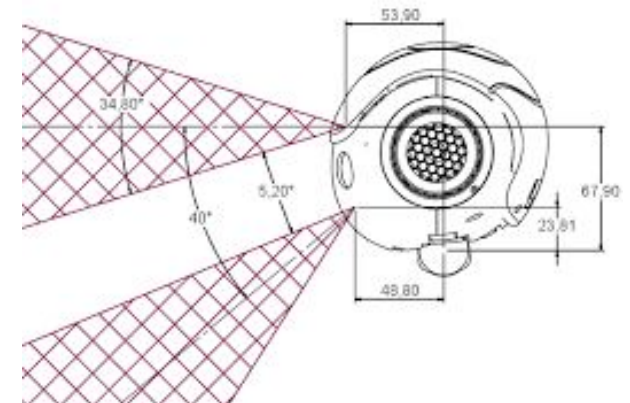


Task & NAO in detail (1 / 3)

- Recognize an object (*ball*) and focus on (*move head appropriate*)
- Pick Up Motion
- Show the object (*ball*)
- Choregraphe (*animation mode*)

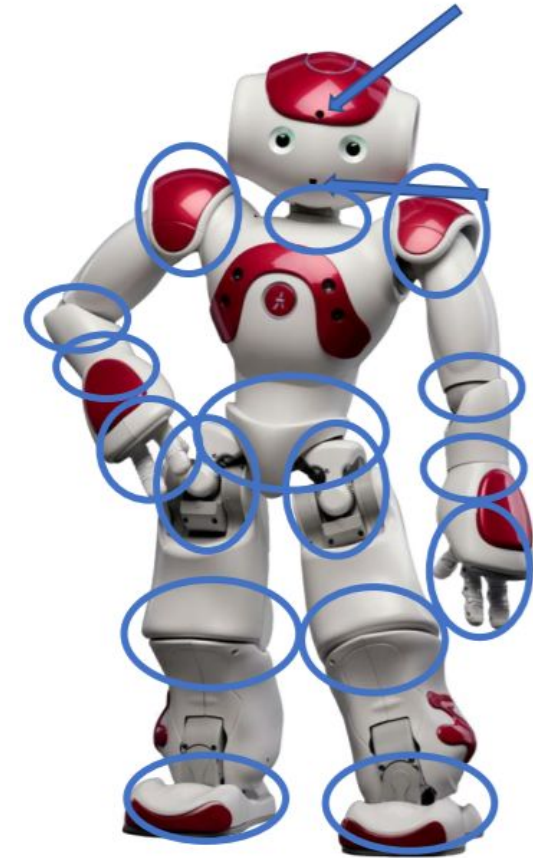
Task & NAO in detail (2/3)

- ❑ **Humanoid robot**
 - ❑ 58 cm tall
 - ❑ Understands and speaks more than 20 languages
- ❑ Equipped with **omnidirectional microphones, speakers, (...)**
- ❑ Additionally equipped with **two 2D cameras**
 - ❑ Able to recognize shapes, objects and even people



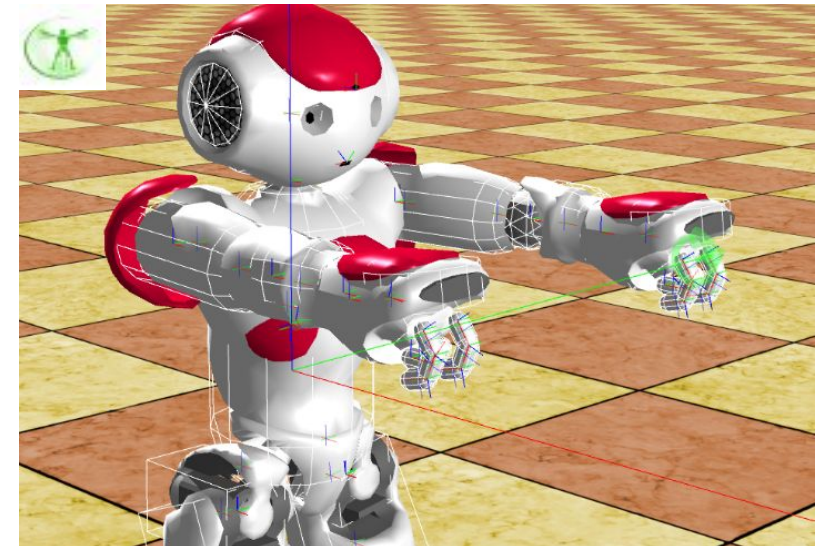
Task & NAO in detail (3/3)

- ❑ Two identical video cameras
 - ❑ Up to 1280x960@30fps each
 - ❑ Top (*id 0*, *AL::kTopCamera*)
 - ❑ Bottom (*id 1*, *AL::kBottomCamera*)
- ❑ 25/26 Joints
 - ❑ LHipYawPitch and RHipYawPitch are connected
- ❑ Programming using Python 2.7 with Jupyter
 - ❑ Library used: NAOqi



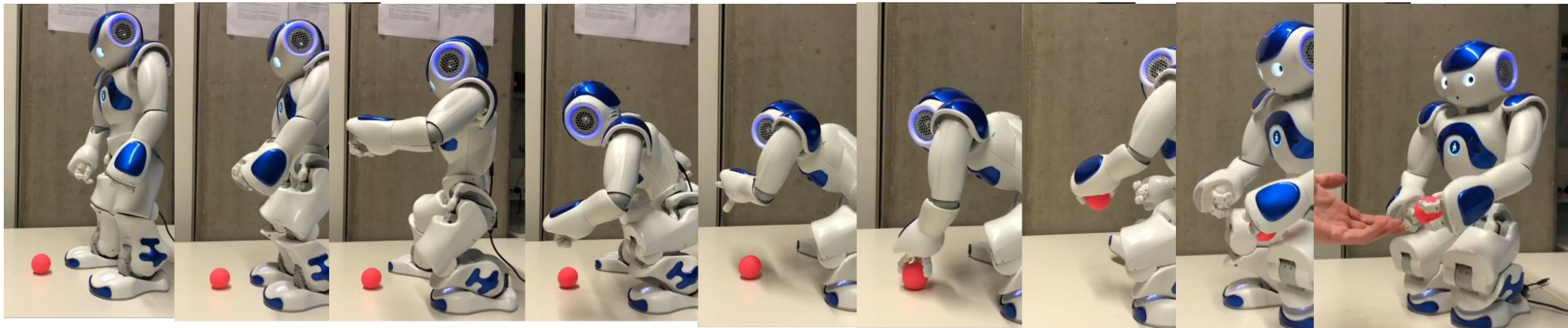
Object detection & interaction (1/7)

- ❑ No behavior in Choregraphe to bend down or pick something up
- ❑ Pick Up Motion steps
 - ❑ Use animation mode
 - ❑ Move joints in desired positions
 - ❑ Print joint positions
 - ❑ Put all positions into an array
 - ❑ Interpolate movement between positions



Object detection & interaction (2/7)

- ❑ No behavior in Choregraphe to bend down or pick something up



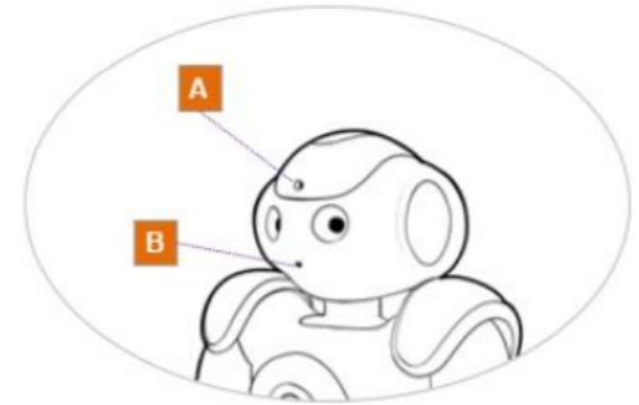
Object detection & interaction (3/7)

- ❑ OpenCV - **open**-source **computer vision** applications
 - ❑ Used for Ball Detection
 - ❑ Offers plenty of useful features, face detection (...)
 - ❑ Subtopic split into three main tasks
 - ❑ Read out images from NAOqi
 - ❑ Detect object by image processing
 - ❑ Following target object with detection as well as movements



Object detection & interaction (4/7)

- ❑ Retrieving the images
 - ❑ Using the two cameras by NAOqi
 - ❑ Each camera is running in a separate thread
 - ❑ NAOqi offers single picture review
 - ❑ Several problems occur with this method
 - ❑ No video streaming function
 - ❑ Only picture-for-picture
 - ❑ FPS rate is low
 - ❑ Internal framework error
 - ❑ In order to reduce resource consumption, events are only fired when a new image was fetched



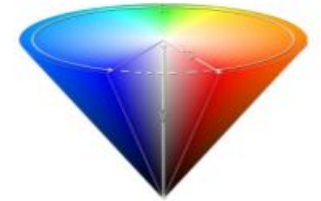
Object detection & interaction (5/7)

❑ OpenCV steps to detect an object

- I. Smoothing (*called blurring*) an image
- II. Convert the image to HSV color space

>> reduce noise

>> get color coords



- III. Create a mask to hold all areas of the defined color

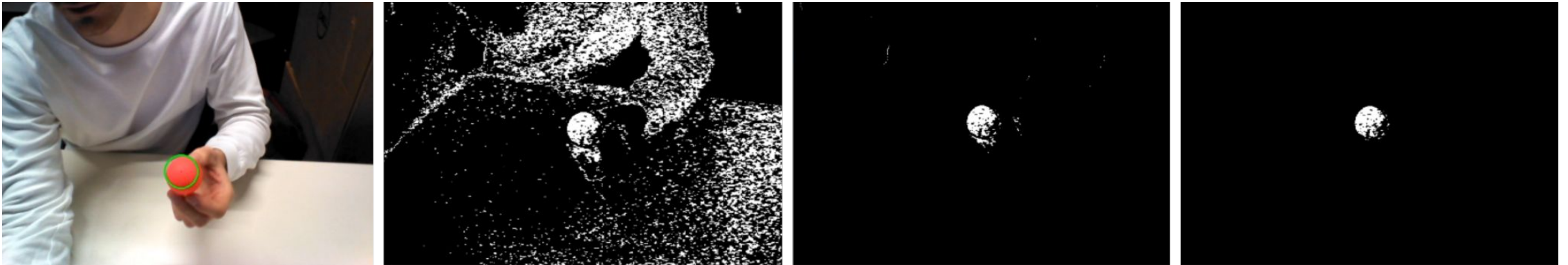
>> get object coords at image

- IV. Draw contours around the object

>> visualize detection

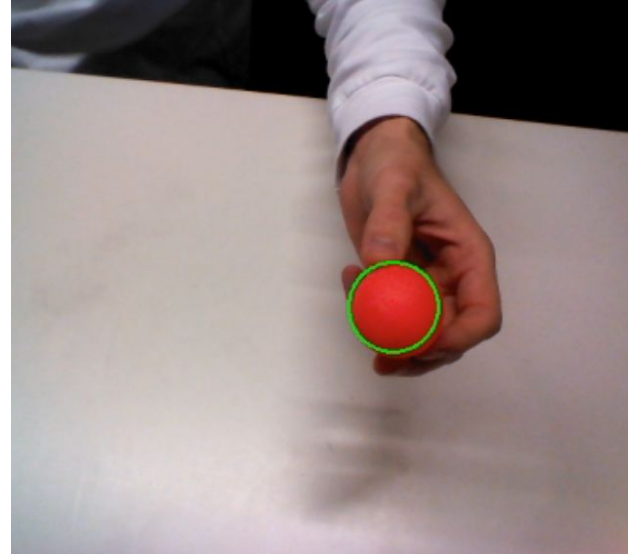
Object detection & interaction (6/7)

- ❑ Initialization of the color



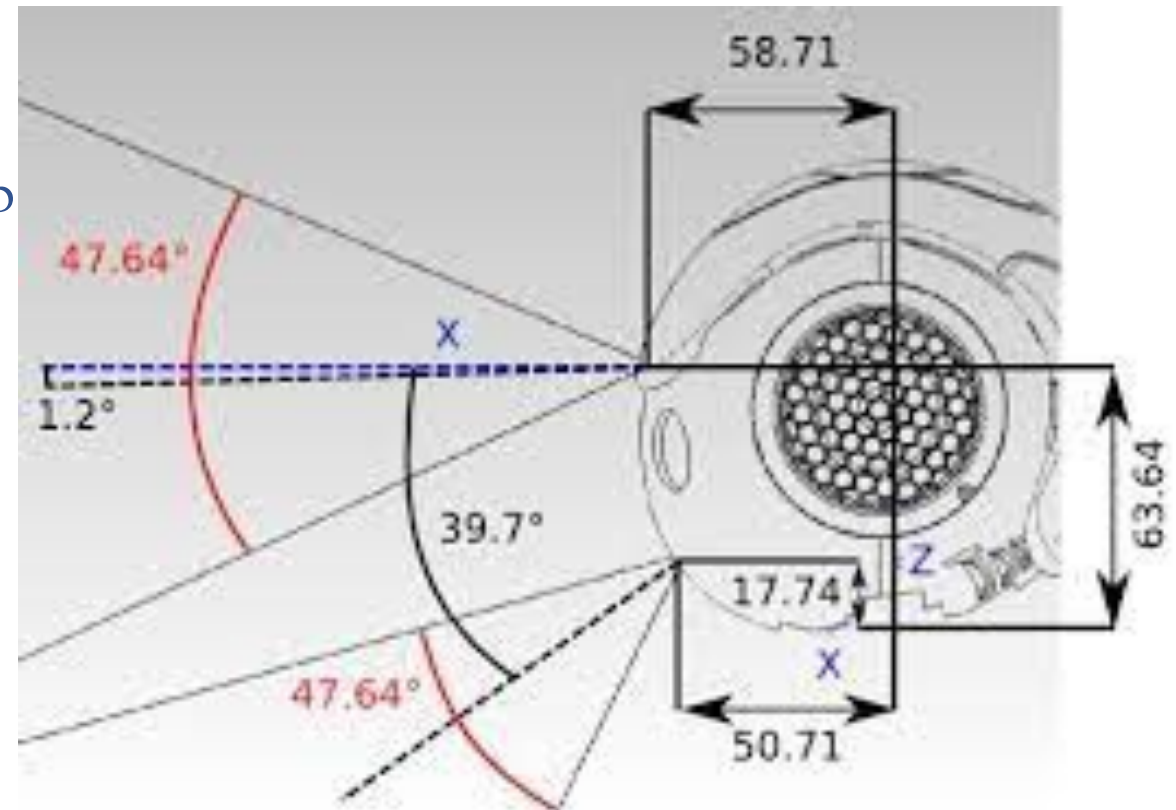
Object detection & interaction (7/7)

- ❑ Detected object
 - ❑ Move head angles to follow the target object
 - ❑ Fire an event, when the ball reaches the pick-up-position



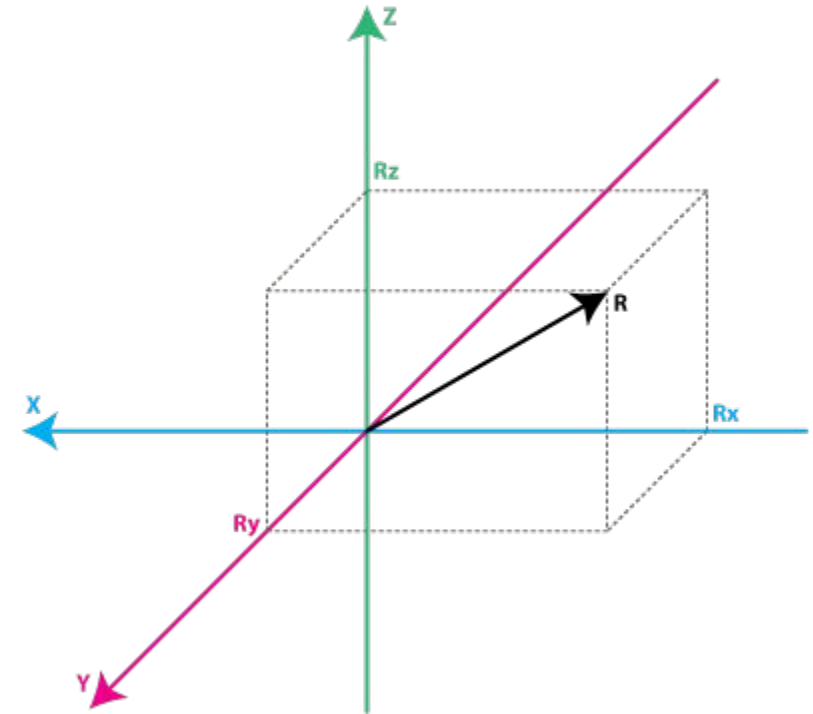
Problems (1/2)

- ❑ NAO has two cameras
 - ❑ **Top:** Scans the horizon
 - ❑ **Bottom:** Scans the lower ground
 - ❑ **Problem:** Both have a narrow overlap of their field of view

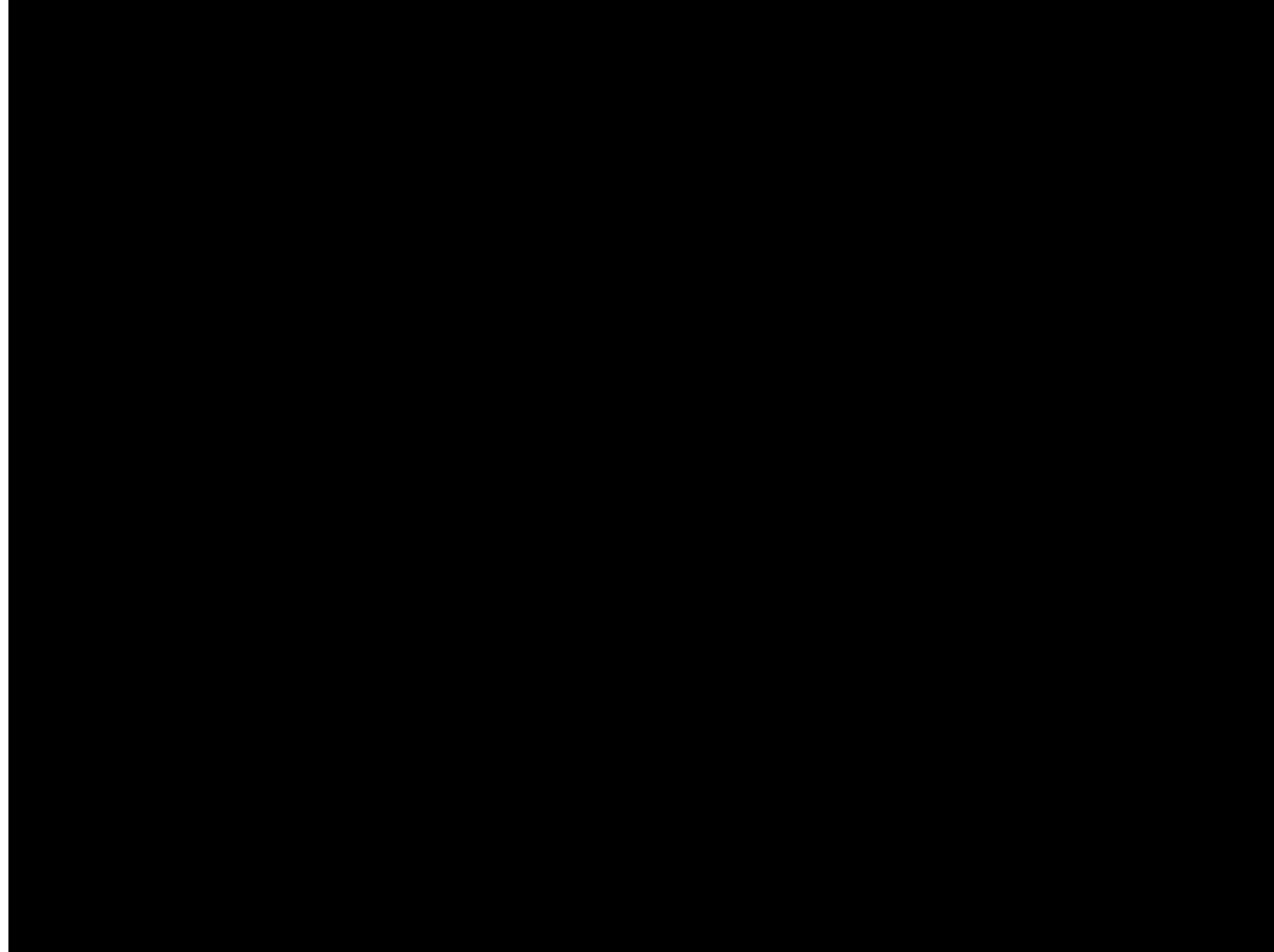


Problems (2/2)

- ❑ Balancing NAO is difficult
 - ❑ **Problem:** No appropriate sensors available for balancing (z-gyroscope not available yet)
 - ❑ A gyroscope X, Y - however - is integrated



Result



Thank you for your attention!
Have any questions? - feel free to ask :)

